

RAJASIMHA MERUGUMALLA

Machine Learning / AI Engineer

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PROFESSIONAL SUMMARY

Computer Science (Data Science) graduate with hands-on experience building production-style AI and machine learning systems, including multi-agent LLM workflows, retrieval-augmented generation (RAG) pipelines, and real-time predictive models. Proficient in Python, SQL, LangGraph, FastAPI, Docker, and gradient boosting frameworks, with a strong foundation in data analytics, ETL, and business intelligence (Power BI, Tableau). Seeking an entry-level ML / AI Engineer role to deliver reliable, measurable, data-driven solutions.

TECHNICAL SKILLS

Programming: Python, SQL

Machine Learning: Regression, Classification, Clustering, Feature Engineering, Model Evaluation, Ensemble Methods, XGBoost, LightGBM, CatBoost, Scikit-learn

LLM & GenAI: LangGraph, Multi-Agent Systems, RAG, Qdrant Vector Database, BM25, Cross-Encoder Reranking, Ollama

Data & Analytics: Pandas, NumPy, Exploratory Data Analysis (EDA), Data Cleaning, Statistical Analysis, ETL Pipelines, PostgreSQL, Star Schema Modeling, A/B Testing Concepts

BI & Visualization: Power BI (DAX, Power Query), Tableau, Excel (Pivot Tables, VLOOKUP), Matplotlib, Seaborn

Tools & Deployment: FastAPI, Docker, Streamlit, Git/GitHub, Jupyter Notebook, VS Code

PROJECTS

Autonomous Multi-Agent Coder (Autocoder) / Personal Project

[GitHub](#)

Tech Stack: Python, LangGraph, Ollama (Qwen2.5-Coder), Docker, Ruff, Bandit

- Architected a local-first multi-agent code generation engine using LangGraph and Qwen2.5-Coder via Ollama, automating repository inspection and code synthesis to reduce manual debugging effort.
- Built a secure Docker execution sandbox with isolated networking (--network=none) and a 512 MB memory limit to safely test untrusted AI-generated Python code.
- Implemented an automated self-healing loop using Ruff and Bandit to detect syntax and security issues and patch them before human Git approval.
- Designed a model fallback router that cascades from local Ollama models to cloud LLM APIs during resource bottlenecks, improving pipeline availability.

RAGLens: Citation-Aware Document Intelligence / Personal Project

[GitHub](#)

Tech Stack: Python, FastAPI, Qdrant, BM25, Cross-Encoders, Reciprocal Rank Fusion

- Built a research-paper RAG pipeline combining Qdrant dense embeddings with BM25 sparse search via Reciprocal Rank Fusion (RRF, k=60), improving context retrieval accuracy by 38%.
- Integrated an ms-marco-MiniLM-L6-v2 cross-encoder for top-k reranking, filtering irrelevant chunks and reducing downstream LLM token usage by 25%.
- Developed a custom CitationValidator that cross-checks LLM-cited IDs against retrieved context metadata, eliminating hallucinated citations before UI rendering.
- Implemented multi-pipeline intent routing in FastAPI for technical Q&A, evidence extraction, and side-by-side literature comparison.

TransitGuard: Predictive Delivery ML Pipeline / Personal Project

[GitHub](#)

Tech Stack: Python, LightGBM, PostgreSQL, FastAPI, Docker, Streamlit

- Built an end-to-end delivery cancellation and delay prediction pipeline with LightGBM, serving real-time risk scores to help mitigate supply chain disruptions.
- Addressed severe class imbalance (4.3% cancellation rate) using cost-sensitive learning (class_weight="balanced"), achieving 97% recall on high-risk cancellations.
- Designed a PostgreSQL data warehouse with star-schema modeling and removed post-checkout data leakage to preserve feature integrity.
- Containerized prediction endpoints with FastAPI and Docker, achieving sub-50 ms inference latency, and built an interactive Streamlit monitoring dashboard.

EDUCATION

Malla Reddy College of Engineering & Technology

B.Tech in Computer Science (Data Science)

Hyderabad, India

2022 – 2026